Super-Resolution of Radargrams with a Generative Deep Learning Model

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***Abstract*—Radar sounder profiles are essential for imaging the subsurface of planetary bodies and the Earth as they provide valuable geological insights. However, the limited availability of high-resolution radargrams poses challenges. This paper proposes a novel method based on generative models to super-resolve radargrams. Our approach addresses the ill-posed and ill-conditioned nature of the super-resolution problem by training a neural network to learn the correlation between radargrams at different scales. The network learns a proxy for the mapping function between ambiguous low-resolution radargrams and more detailed high-resolution ones, considering the data's geological and statistical properties. The mapping function enables the super-resolution of previously unseen low-resolution radargrams acquired in comparable conditions to those in the training and imaging of similar underlying geology. To achieve this, we adopt a generative Cycle-Consistent Adversarial Network (CycleGAN), explicitly designed to match properties between low- and high-resolution radargrams, accounting for variations in dimensions and radiometric properties. Furthermore, we enhance the network performance by incorporating skip connections, a ResNet module, and attention mechanisms. The proposed method is validated using MCoRDS3 radargrams acquired in Greenland and Antarctica as high-resolution data. As low-resolution data, we used simulated radargrams representing what is expected by an Earth-orbiting low-resolution radar sounder to have a controlled experiment. The results are evaluated qualitatively and quantitatively, focusing on the areas with reflections with complex shapes that may generate artifacts and unrealistic geological features.**

***Index Terms*—Super-resolution, Deep Learning, Image Processing, Radar Sounder.**

I. Introduction

RADAR sounders (RSs) are increasingly used for Earth observation and planetary exploration as they image the subsurface from remote. RSs generate radargrams, which are profiles of the subsurface that collect the backscattered echoes from the buried dielectric, thermal, and mechanical interfaces [1]. Radargram analysis is crucial to improving our understanding of the geologic structures and processes in the terrestrial [2, 3] and planetary subsurface [4, 5]. However, geological target detection and characterization are limited by the relation between the target size and the radargram resolutions in the range and azimuth directions [6, 7]. The azimuth resolution depends on many factors, including the antenna pattern, the acquisition geometry, and the surface roughness. It can be approximated as the antenna footprint size in the along-track direction [7]. Most common RS antennas are characterized by a broad radiation pattern and a large ground footprint, resulting in a poor azimuth resolution [7]. Approaches to improve the azimuth resolution include exploiting the large antenna footprint and Doppler shift due to the relative motion between the platform and the target to generate a longer synthetic antenna with SAR-focusing algorithms [7]. A large antenna footprint indicates a higher amount of undesired echoes backscattered from off-nadir that may mask the subsurface signal.

The range resolution is inversely proportional to the bandwidth, so a finer resolution requires a wider bandwidth. A wide bandwidth can be obtained by transmitting signals linearly modulated in frequency (i.e., chirp pulses) [7] and exploiting pulse compression in the time domain to increase the resolution. However, RSs that deeply penetrate the subsurface require low central frequencies, leading to low bandwidth and consequently low range resolution due to hardware limitations related to the fractional bandwidth (i.e., the ratio between the bandwidth and the central frequency). Super-resolution methods that partially overcome Rayleigh's criterion use the BandWidth Extrapolation (BWE) technique to enhance the range resolution only [8, 9]. BWE techniques such as [8] use a linear prediction model that cannot accurately generalize complex rough surfaces.

This paper proposes a novel method based on a generative adversarial model to super-resolve radargrams. The method, which extends and develops the initial idea in [10], proposes to solve the ill-posed and ill-conditioned super-resolution problem by training a generative model to learn the relation between the low- and high-resolution radargrams in a specific subsurface scenario. During the training, we inject information into the neural network about the relation between the geometric and radiometric structures in high- and low-resolution data. This teaches the network to super-resolve previously unseen low-resolution data based on the geologic and statistical features learned during the training. As a generative network, we adopted a cycle-consistent generative adversarial network (CycleGAN) that can be trained unsupervised and does not require pairs of high- and low-resolution patches sharing the same ground track. We propose adapting the CycleGAN architecture from the literature to consider the properties of low— and high-resolution radargrams and the super-resolution problem. Moreover, we include a feature-reconstruction loss in the training to sharpen the geometrical details. The proposed method has been successfully validated in a controlled condition where the high-resolution data are MCoRDS3 airborne radargrams acquired in Greenland and

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Antarctica. Low-resolution data are simulated from MCoRDS3 airborne radargrams with an analog simulator [6] to match the properties of a hypothetical Earth-orbiting radar sounder. The results have been evaluated qualitatively and quantitatively, focusing on areas (ice stratigraphy and basal interface) with complex reflections to better understand the impact of artifacts.

The structure of the paper is as follows. Section II defines the radar resolutions and illustrates the literature overview on remote sensing super-resolution. Section III introduces the super-resolution problem as an inverse ill-posed and ill-conditioned problem. Section IV defines the proposed method for super-resolving radargrams. Section V describes the datasets, and Section VI discusses the experimental results that show the effectiveness of the proposed method. Section VII discusses the potentials and limitations of the proposed method and concludes the paper.

II. Background

A. Background on radar resolutions

The azimuth resolution ρ_a is normally identified as the first Fresnel zone on the surface and depends on many factors, such as the antenna pattern, the platform height, and the surface roughness [7]. After Doppler focusing, the azimuth resolution is approximated as $\rho_a \approx V_S/B_D$, where V_S is the moving platform velocity. B_D is the Doppler bandwidth that is defined as:

$$B_D = \frac{2V_S^2}{h\lambda}T_i. \tag{1}$$

where T_i is the integration time, defined by the time the antenna illuminates the target. In the maximum theoretical case, $T_i = L_S/V_S$, where L_S is the synthetic antenna length [7]. In the case of a flat surface, the coherent scattering from the ground is commonly limited to the first Fresnel zone [7]. The diameter of the first Fresnel zone is given by $D_F \approx \sqrt{2\lambda h}$ (focused SAR processing [7]), where h is the platform height, and λ is the signal wavelength. In the case of a rough surface, the scattering is not limited to the Fresnel zone but to the whole antenna aperture, which is normally larger than D_F. For example, for a dipole antenna, the footprint diameter is defined as $D_A \approx 0.88\frac{h\lambda}{L_A}$, where L_A is the physical length of the dipole. The larger footprint comes with a higher amount of undesired reflections from off-nadir (clutter) that may mask the subsurface signal and reduce the signal-to-noise ratio (SNR). Decreasing the clutter requires a more directive antenna and, thus, a narrower footprint that would degrade the azimuth resolution.

The range resolution ρ_r depends on the central frequency and the bandwidth [7]. The maximum penetration depth depends on many factors related to the radar system parameters (mainly the RS central frequency), the signal propagation loss above the surface and in the subsurface (spreading and absorption losses), the noise added by the receiver, and the cosmic background noise (only for spacecraft-borne RS) [11]. Considering a fixed propagation loss, the main radar parameter influencing the penetration is the signal's central frequency f_c– a decrease of f_c leads to an increase of the maximum penetration [7].

Considering linearly modulated signals (chirps), the range resolution (ρ_r) mostly depends on the propagation medium and the chirp signal bandwidth B [7]:

$$\rho_r = \frac{c}{2B\sqrt{\varepsilon_r}} \tag{2}$$

where c is the speed of light and ε_r is the relative dielectric permittivity of the propagation medium.

B. Related work on remote sensing super-resolution

The main method for super-resolving RS data is BWE. BWE technique assumes that the total signal scattered by a point is a superimposition of complex sinusoids, which is valid in the case of a point scatter inside a non-dispersive and lossless volume [12]. Under this assumption, the Auto-Regressive (AR) spectral estimator can extract the sinusoids over the original bandwidth and expand the signal over a wider bandwidth in the transformed Fourier domain. The AR performs a linear prediction by considering a combination of past and future signals, and the unknown coefficients are estimated by minimizing the error between the predicted and the actual signal. BWE-based methods were applied to RS data with promising results on simulated data, MARSIS/MarsExpress data [9, 13], SHARAD/MRO data [14, 13], WISDOM/ExoMars radar [12], and SELENE (Kaguya) Lunar Radar Sounder data [15]. However, when the method assumptions are not satisfied (e.g., distributed surface), BWE introduces several artifacts (e.g., line splitting and undesired reflections) [8]. The super-resolved radargrams quality depends on i) the generalization capability of the model built with a limited number of samples (ill-posed and conditioned problem); ii) the systematic errors due to the model assumptions, including the propagation effects; and iii) the low signal-to-noise ratio of the data. Since the method syntheses a larger signal bandwidth, it enhances the range resolution only but not the azimuth resolution. Finally, since the method super-resolves each rangeline separately, the super-resolved reflections may be discontinuous. This can be overcome using deep-learning networks embedding filters considering spatial context information.

The remote sensing literature has proposed several super-resolution methods for SAR and optical Earth surface imagery based on machine learning that simultaneously enhance the resolutions in longitude and latitude. They consider prior information, e.g., based on the spatiospectral fusion and multi/hyperspectral image fusion [16, 17].

Recently, these methods have been outperformed by deep learning (DL)-based approaches [18]. The most important advantage of DL is the ability to manage data at different processing levels (e.g., L1, L2), mitigate the need for manual feature extraction, and unravel complex nonlinear mapping between the input and output, at the cost of large and diverse training datasets [18]. DL architectures are trained to learn the nonlinear mapping function between low- and high-resolution data and are based on a combination of well-known elements in the computer vision literature, including i) upsampling layers that interpolate the low-resolution features to generate higher-resolution ones [19, 20], ii) residual learning that exploits the difference (residual) between the input

low-resolution data and the expected high-resolution output, exploiting the high correlation between the input and output data [21, 22], iii) Laplacian pyramid structure that extracts features at different scales based on the Gaussian pyramid structure [23], and iv) attention mechanism that focuses the network on the most important details of the features [24].

Super-resolution methods for radar images are challenged due to the multiplicative noise (speckle) and the loss of information given by the data characterized by fewer channels than those considered in computer vision and optical remote sensing imagery [25]. Therefore, super-resolution methods for optical data cannot be applied to radar data (SAR, GPR, and RS) that require the designing of ad-hoc methodologies. State-of-the-art super-resolution methods based on deep learning for radar data focus on previously despeckled SAR imagery [26, 27, 28, 29]. Wang et al. [26] proposed to downsample despeckled SAR images with a Gaussian kernel (blurring operation) followed by the reconstruction to the original resolution with a generative adversarial network (GAN). Authors in [27] propose to use a GAN that preserves high-frequency details and is trained with despeckled SAR imagery simulated from optical data. For despeckled polarimetric SAR data, authors proposed deep architectures for super-resolution that exploit the different information in the polarimetric channels [28, 29]. Few works process no-despeckled SAR imagery with an end-to-end deep solution for super-resolution [30, 31, 32]. Yu et al. [30] propose a weighted dense connected convolutional network adapted from the optical remote sensing literature and trained on synthetic SAR data. The super-resolution problem is unrealistically modeled as done for the optical remote sensing images (blurring, downsampling, and addition of additive white Gaussian noise). This super-resolution model is not valid for radar data given the high dependence of the radargram spatial resolution with the instrument parameters, acquisition geometry, and sensed environment (see Section II). The results on the simulated data are radiometrically unrealistic as the connection between the data resolutions and the instrument parameters is not considered. Cen et al. [31] proposed a multi-Convolutional Neural Network (CNN) architecture with three modules trained in a supervised manner on bistatic radar data to extract features at different resolutions. Helal-Kelany et al. [32] proposed to train a scale-invariant residual network with amplitude and phase signals to coregister two single-look complex InSAR images acquired at different resolutions. The network training is supervised and exploits the mutual information in the InSAR images to perform coregistration and super-resolution simultaneously. Although these methods achieved good results on SAR data [31, 32], the supervised training is a limitation and requires couples of low- and high-resolution images acquired over the same ground track. Unfortunately, this is not the case for RS data, as there are not enough data sharing the same track to generate a diverse and large dataset for robust supervised training of a deep learning network. To summarize, a need emerges for novel radargram super-resolving methods that solve the inverse problem and simultaneously enhance the azimuth and the range resolutions, considering RS data-specific statistical properties, including the noisy components.

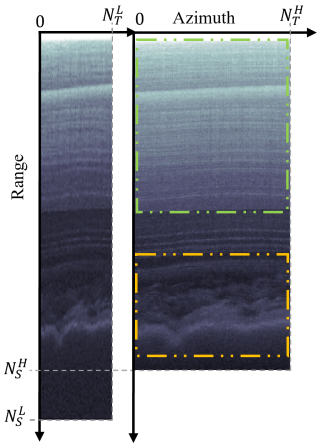


Fig. 1: Low- and high-resolution radargrams acquired by sensors with different parameters. A lower frequency implies lower bandwidth and lower range and azimuth resolutions (left), while a higher frequency allows a wider bandwidth and a higher range and azimuth resolutions (right). The yellow and green boxes highlight regions of interest for the local analysis.

III. Problem Formulation

Let us consider two radargrams $\mathbf{R_L}$ and $\mathbf{R_H}$ belonging to the low-resolution $\mathcal{L}$ and the high-resolution $\mathcal{H}$ domains, respectively. $\mathbf{R_L}$ and $\mathbf{R_H}$ are acquired by two RSs with different system parameters so that the range and azimuth resolutions of $\mathbf{R_L}$ are lower than those of $\mathbf{R_H}$. Each radargram can be defined as a 2D matrix of N_T^i traces (columns) and N_S^i samples (rows):

$$\mathbf{R}_i = \{R_i(r,a) | a \in [1, N_T^i], r \in [1, N_S^i]\} \tag{3}$$

where $i = [L, H]$, R_i is the power of the reflections in the radargrams, and (a, r) are the azimuth and range coordinates, respectively. Adjacent traces capture information on the adjacent vertical portions of the subsurface and image adjacent geologic targets. Given the different RS parameters, the range and azimuth resolution of data and the penetration depth varies so that the number of traces N_T^L and N_T^H and samples N_S^L and N_S^H of the two radargrams covering the same along-track length are different (see Fig 1). N_S depends on the RS maximum penetration depth and the range resolution ρ_r so that a larger value of N_S is related to a finer range resolution and a deeper maximum penetration (see Fig. 1).

This paper aims to enhance both the range and the azimuth resolution of radargrams so that the resolutions of $\mathbf{R_L}$ reach those of $\mathbf{R_H}$. The super-resolution is defined as an inverse problem [18]. Mathematically, the forward problem is defined as [33]:

$$\mathbf{R_L} = \mathcal{F}(\mathbf{R_H}) \tag{4}$$

where $\mathcal{F}(\cdot)$ is the function that approximates the forward problem and maps $\mathbf{R_H}$ into $\mathbf{R_L}$. Consequently, the inverse problem can be described as:

$$\mathbf{R_H} = \mathcal{F}^{-1}(\mathbf{R_L}) \tag{5}$$

where $\mathcal{F}^{-1}(\cdot)$ is the inverse function that describes the super-resolution and solves the inverse problem. Thus, the inverse

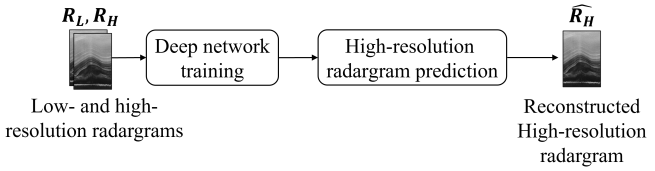


Fig. 2: Flow chart of the proposed super-resolution method.

problem can be described as the estimation of the hidden parameters of $\mathcal{F}^{-1}(\cdot)$. An inverse problem is well-defined if: i) for any $\mathbf{R_L}$ there is a $\mathbf{R_H}$ that satisfies (4) (property of existence); ii) for every $\mathbf{R_L}$ there is only one $\mathbf{R_H}$ that solves the problem (property of uniqueness), and iii) $\mathcal{F}^{-1}(\cdot)$ is a continuous map so that small changes in $\mathbf{R_L}$ correspond to small changes in the estimated $\mathbf{R_H}$ (property of stability) [33]. However, in the super-resolution inverse problem, the uniqueness and stability properties are not satisfied. Uniqueness is not satisfied as the number of parameters to estimate is much larger than the number of known parameters, and the inverse model depends on unknown observables. Moreover, the number of pixels to estimate in $\mathbf{R_H}$ are more than those in $\mathbf{R_L}$: given two radargrams with the same penetration and same length, this can be mathematically rewritten as $N_T^L << N_T^H$ and $N_S^L << N_S^H$. The subpixel information required for super-resolving the radargrams is not explicit in the low-resolution data. The stability property is not satisfied as small variations of $\mathbf{R_L}$ do not correspond to small variations of $\mathbf{R_H}$.

IV. Proposed Super-Resolution Method

The proposed super-resolution method exploits generative deep-learning models to synthesize high-resolution data. The generative model is trained to learn the function $\mathcal{G}(\cdot) \approx \mathcal{F}^{-1}(\cdot)$ to map the distributions of the input in $\mathcal{L}$ to the output in $\mathcal{H}$. Learning $\mathcal{G}(\cdot)$ teaches the network to extract meaningful features for super-resolving previously unseen low-resolution data acquired in conditions similar to those of the training. The inverse problem is solved by injecting prior information correlating low- and high-resolution data into the network.

The method consists of two main steps (see Figure 2). First, the network is trained to learn i) the spatial and radiometric/statistical properties of the RS data in the $\mathcal{L}$ and $\mathcal{H}$ domains, and ii) the mapping function $\mathcal{G}(\cdot)$ that approximates $\mathcal{F}^{-1}(\cdot)$. The training set consists of low- and high-resolution data acquired over homogeneous geological areas (i.e., imaging similar geologic subsurface targets). The standard training is enriched with the feature reconstruction loss in [34] to mitigate the low-pass effect and generate super-resolved patches. As a deep generative architecture, we propose using a CycleGAN that can be trained unsupervised [35]. We design a novel generator architecture considering the different low- and high-resolution data resolutions and their dimensions. To boost the performance as a generator, we consider a U-shaped CNN enriched with skip connections, attention gates, and a residual learning module. Secondly, the mapping function $\mathcal{G} \approx \mathcal{F}^{-1}(\cdot)$ is used to accurately super-resolve previously unseen radargrams acquired with the same sensor and imaging similar geology to those of the training data.

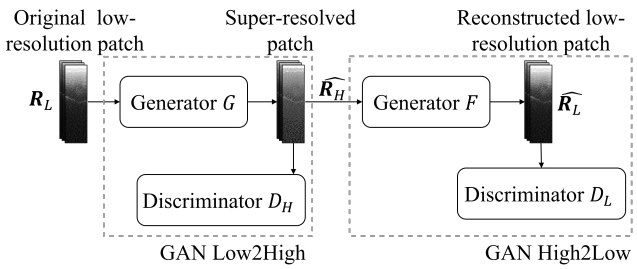


Fig. 3: Diagram of the CycleGAN training.

A. Architecture of the Deep Network

A CycleGAN consists of two GANs (called Low2High and High2Low) (see Fig. 3), each made of two networks (a generator and a discriminator) that are simultaneously trained. During the training, the generators learn the geometric and radiometric properties of low- and high-resolution data to extract meaningful features and generate accurate super-resolved images that fool the discriminator. As Fig. 3 shows, the first GAN (Low2High) learns the inverse mapping function $\mathcal{G}(\cdot)$ that accurately super-resolves $\mathbf{R}_L$ into $\widehat{\mathbf{R}}_H$. The second GAN (High2Low) learns the direct mapping function $\mathcal{F}(\cdot)$ that recovers $\mathbf{R}_L$ from $\widehat{\mathbf{R}}_H$. In the following, the structure of the CycleGAN architecture is described in detail.

1) Architecture of the Generators: As generators, we consider two networks with a U-shape– (i) the first part encodes the input into a lower dimension (latent space), and ii) the second part decodes the latent features (see fig. 4). The Low2High generator provides as outputs the super-resolved images, and the High2Low generator provides the reconstructed low-resolution ones. Let us define the input and output of each network layer as $H_s^{l,j}$, where $l \in [1,3]$ indicates the encoder and decoder block number, j can be either $\mathcal{E}$ or $\mathcal{D}$ and indicates the encoder and the decoder blocks, respectively. $H_s^{l,j}$ are the features extracted by the l-th encoder or decoder block and have dimensions $(M^l \times h^l \times w^l \times M_0)$, where M^l indicates the number of features, w^l and h^l are their horizontal and vertical dimensions, respectively, and M_0 is the input patch channel size that is fixed to $M_0 = 1$. The index s indicates the type of processing within the block– $\mathcal{C}$ stands for a convolutional layer, followed by instance normalization [36] and ReLU activation function layers, $\mathcal{U}$ for upsampling layer, and $\mathcal{V}$ for the dropout. Finally, H_R is the output of the residual module and has shape $(M^R \times h^R \times w^R)$, where M^R indicates the number of features, and h^R and w^R the height and the width of the features, respectively. The residual module does not change the dimension of the input data– for example, in Fig. 4, $h^R = h^3$ and $w^R = w^3$. H_{AG}^l is the output of the l-th attention gate (i.e., the input of the l-th decoder block) and has shape of $(M^{l+1} \times h^{l+1} \times w^{l+1})$.

Encoder. The encoder comprises N_L downsampling blocks, consisting of a convolution layer, followed by an instance normalization layer [36], and a rectified linear unit (ReLU)

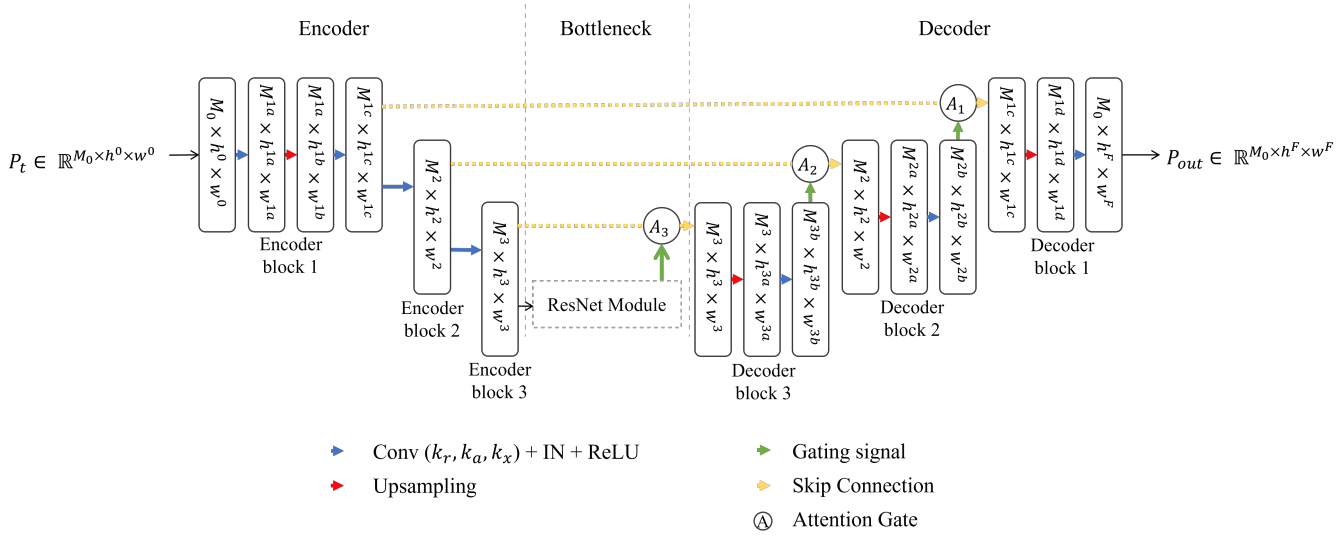


Fig. 4: Architecture of the generators based on a U-Net embedded with the ResNet and attention modules.

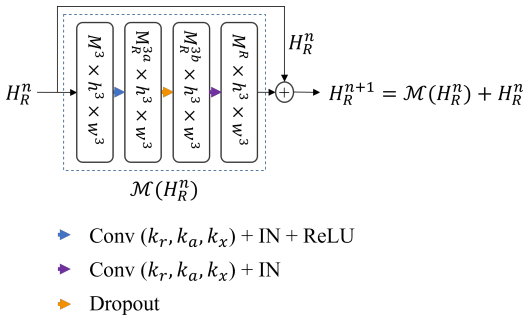


Fig. 5: Structure of one ResNet block repeated N_R times to form the ResNet module at the bottleneck. The layers surrounded by the dashed lines approximate the residual function $\mathcal{M}(H_R^n)$ that is added to the input H_R^n to fit the desired function $\mathcal{N}(H_R^n)$ and extract the output features H_R^{n+1}.

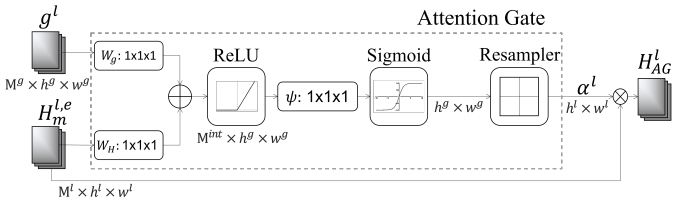


Fig. 6: Attention Gate module structure to filter the signal from the skip connection $H_C^{l,\mathcal{E}}$ [37]. The gate signal g^l highlights the salient spatial regions of the feature set $H_C^{l,\mathcal{E}}$ and determines the attention coefficients α^l. The attention coefficients are then element-wise multiplied with the low-level feature set to obtain the output of the l-th AG module H_{AG}^l.

introducing nonlinearity (see Fig 4). Each convolutional layer l learns high-level features from the spatial and radiometric information in the data. The input of the first block $(l = 1)$ is a radargram patch $\mathbf{P} = P_t, t \in [1, N_t]$. Each block l provides as output a set of M^l features. For extracting the features, each convolutional layer l applies a convolution across the input with a kernel of size (k_r^l, k_a^l), such that the learnable parameters set $W^l \in \mathbb{R}^{M^l \times k_r^l \times k_a^l \times M_0}$. M_0 is related to the input radargram and fixed to $M_0 = 1$. The convolutional layer output is processed with instance normalization that normalizes a mini-batch of data for each observation independently to improve the training convergence and reduce the sensitivity to hyperparameters [36].

Residual module. In the network bottleneck, a residual connection module extracts meaningful deep features exploiting shortcut connections to mitigate the problems of vanishing/exploding gradient and degradation (saturation of the accuracy) [21]. Residual learning compels the network to capture the residual, or the difference, between the input low-resolution data and the desired high-resolution output by utilizing the strong correlation between the input and output data considering the global [22] and local [21] context. The ResNet module comprises N_R concatenated residual blocks, each extracting a set of features $H_R^n, n \in [1, N_R]$. The input of the ResNet module is the set of features extracted from the last block of the encoder $(H_c^{l=N,\mathcal{E}})$. The last residual block $(n = N_R)$ extracts the features H_R fed to the decoder. Given an input H_R^n, let us define $\mathcal{N}(H_R^n)$ as the mapping function desired to be fit by the staked layers. The core idea of residual learning is to force the staked layers to asymptotically approximate the residual mapping function $\mathcal{M}(H_R^n) := \mathcal{N}(H_R^n) - H_R^n$ that is easier to optimize than $\mathcal{N}(H_R^n)$ [21]. He et al. [21] proposed an implementation with stacked nonlinear layers connected with shortcuts that perform the identity mapping. Therefore, the features from the shortcuts (H_R^n) are added to the output of the stacked layers fitting $\mathcal{M}(H_R^n)$ (see Fig. 5). In this work, each residual block consists of a convolutional layer followed by instance normalization and ReLu activation layers, a dropout layer, and another convolutional layer followed by an instance normalization layer [36]. The output is the residual $\mathcal{M}(H_R^n)$, which is added to the input (H_R^n) from the shortcut.

Decoder and Attention Gates (AGs). The decoder consists

of N_L expanding blocks with the attention gate in [37] to filter the noisy features from the skip connections and activate the most relevant regions only. The inputs of the l-th attention gate are the features from the next lowest layer $(l+1)$ as gating signal g^l and the features from the skip connections ($H_{\mathcal{C}}^{l,\mathcal{E}}$). For the $l(=N_L)$-th AG, the gating signal consists of the output features of the ResNet module H_R (see the green arrows in Fig. 4), i.e., $g^l = H_R$. For the other AGs ($l \in [1, N_L-1]$), the gating signal is the output of the previous layer $(l+1)$, i.e., $g^l = H_{\mathcal{C}}^{l+1,\mathcal{D}}$ and $M_g = M^{l+1}$. Although skip connections help to prevent the vanishing gradient problem and to propagate the spatial properties, features $H_{\mathcal{C}}^{l,\mathcal{E}}$ are redundant and poorly representative of the radar data. Given that they are extracted at a higher level, features from the decoder better generalize the data. Hence, decoder features can be used as a gating signal to filter $H_{\mathcal{C}}^{l,\mathcal{E}}$ and activate the most informative regions [37]. To this end, the l-th AG module extracts a scalar attention value $\alpha \in [0,1]$ from $H_{\mathcal{C}}^{l,\mathcal{E}}$ based on the gating signal g^l [37]. Each AG applies three linear transformations (W_g, W_H, and Ψ) to g^l and $H_{\mathcal{C}}^{l,\mathcal{E}}$ with channel-wise 1D convolutions (see Fig. 6). W_g and W_H map the input features g^l and $H_{\mathcal{C}}^{l,\mathcal{E}}$ into a $\mathbb{R}^{int}$ space, while Ψ reduces the feature number from M_{int} to 1. The resulting features are summed to obtain the gating coefficients (additive attention), which are processed by a ReLU and a channel-wise 1D convolution that provides one feature per pixel as output and is followed by a sigmoid activation function. The sigmoid output signal is resampled with a bilinear interpolation by an upsampling layer that doubles the dimensions to match the shape of the following decoder block input. This generates a grid signal conditioned to the data spatial information so that the attention coefficient α^l of the l-th attention gate has a $(h^l \times w^l)$ shape. Finally, the output of the l-th attention gate H_{AG}^l is the element-wise multiplication $[\cdot]$ between the features in $H_{\mathcal{C}}^{l,\mathcal{E}}$ and the attention coefficients α^l, i.e., $H_{AG}^l = H_{\mathcal{C}}^{l,\mathcal{E}} \cdot \alpha^l$. We concatenated the l-th AG H_{AG}^l output with the upsampled version of the gating signal $H_{\mathcal{U}}^{l+1,\mathcal{D}}$ to generate $H_{Au}^{l+1,\mathcal{D}}$. Therefore, for the first decoder block ($l = N_L$), the AG output is concatenated with the upsampled feature from the ResNet module H_R. For the other decoders block ($l \in [1, N_L-1]$), the AG output is concatenated with the upsampled features from the previous network layer to generate $H_{Au}^{l+1,\mathcal{D}}$.

Decoder blocks $l \in [1, N_L]$ consist of an upsampling layer followed by a convolutional layer with kernel size (k_r^l, k_a^l) and stride dimension (k_x^l, k_y^l), an instance normalization layer [36], and a ReLU (see Fig. 4). The inputs of each block are the upsampled features from the next lowest block concatenated with the skip connection signal filtered by the attention gate. Each block l provides as outputs M^l features $H_{\mathcal{D}}^l$ of size $(M^l \times h^l \times w^l)$. The last block output P_{out} is the super-resolved image for the Low2High generator and the reconstructed low-pass patch for the High2Low generator, respectively.

2) Architecture of the Discriminators: As discriminators, we consider an architecture similar to those in [35]. The input of the discriminator D_H (see Fig. 3) is the original and the super-resolved high-resolution patch, while discriminator D_L receives in input the original and the reconstructed low-resolution data. Each discriminator consists of five blocks of a convolutional layer that uses the stride operation to halve the input features dimension in the horizontal and vertical directions, followed by an instance normalization layer [36] and a ReLU activation layer. The five blocks are followed by a convolutional and an instance normalization layer. The discriminator output P_D has dimension $M_D \times h_D \times w_D$ with $M_D = 1$, and its values are rounded to the closest integer value, i.e., 0 or 1.

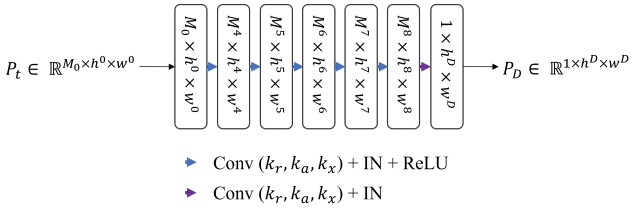


Fig. 7: Architecture of the discriminators.

B. Deep Network Training

To train the network, the input radargrams $\mathbf{R}$ are split into non-overlapping patches $P_t^i, t \in [0, N_P]$ of size (h_0^i, w_0^i). The vertical dimension of the patches is chosen to be the same as the radargram depth ($h_0^i = N_S^i$) so that the samples capture the global context and are fully representative of the subsurface (see Section III) in terms of the variability, pattern, and vertical spatial distribution. In the azimuth direction, the patch width w_0 is a tradeoff between covering a large portion of the subsurface and optimizing the computational load. After the processing, the output patches are concatenated to have the same shape as the original radargrams.

The standard training of the cycleGAN consists of three loss functions to be minimized to optimize the network parameters to extract meaningful features. The three loss functions are two adversarial losses $L_{GAN}^i, i = [L, H]$ to train the two GANs (Low2High and High2Low, respectively) and a cycle-consistency loss L_{cyc} [35]. The adversarial losses train the Low2High and High2Low generators to learn the two mapping functions $\mathcal{G}(\cdot)$ and $\mathcal{F}(\cdot)$ that produce the high-resolution data ($\widehat{\mathbf{R}}_H$) and the reconstructed low-resolution data ($\widehat{\mathbf{R}}_L$), respectively. On the other hand, the cycle-consistency loss simultaneously trains the whole architecture so that the mapping functions $\mathcal{G}(\cdot)$ and $\mathcal{F}(\cdot)$ reverse each other. Accordingly, $\mathcal{G}(\cdot) \approx \mathcal{F}^{-1}(\cdot)$. We further enrich the training with the feature reconstruction loss L_{feat} inspired by [34] to mitigate low pass effects and force the network to learn subsurface targets' geometrical and radiometric details. Considering the i-th GAN, the adversarial loss L_{GAN}^i simultaneously trains the generator and the discriminator. The generator synthesizes realistic images from the original radargram and tries to fool the discriminator. The discriminator, on the other hand, should learn the differences between real and fake data to unveil the fake ones. In the case of the Low2High GAN, the adversarial loss for the generator G that learns the mapping function $\mathcal{G}(\cdot)$

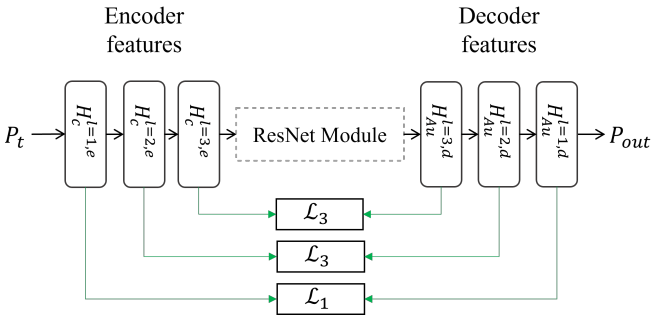


Fig. 8: Flow chart of the generator architecture and the feature reconstruction loss (L_l) for each l pair of encoder-decoder blocks.

TABLE I: Parameters of MCoRDS [3] for the campaigns acquired in Greenland and Antarctica. Parameters of the Earth observation RS (EORS) [38, 6] for simulating the low-resolution datasets. Geometrical resolutions of the radargrams, considering the ice dielectric permittivity equal to $\varepsilon_r = 3.15$. AGL stands for above ground level

Parameters	**Greenland dataset**	**Antarctica dataset**	**EORS dataset**
Central frequency (MHz)	195	190	45
Wavelength (m)	1.54	1.58	6.66
Bandwidth (Mhz)	30	50	10
Sampling Frequency (MHz)	30	50	12
Transmitted Power (kW)	1.050	6	0.2
Altitude AGL (km)	0.5	0.5	500
Range Resolution in ice (m)	4.3	2.6	10
Azimuth Resolution (m)	27.5	27.5	300

is defined as:

$$L_{GAN}^{H}(\mathcal{G}, D_H, \mathcal{L}, \mathcal{H}) = E_{\mathbf{R}_H \sim pdata}(\mathbf{R}_H)[\log D_H(\mathbf{R}_H)] + E_{\mathbf{R}_L \sim pdata}(\mathbf{R}_L)[\log(1 - D_H(\mathcal{G}(\mathbf{R}_L)))] \quad (6)$$

where i) the generator G synthesizes the image $\mathcal{G}(\mathbf{R_L})$ that has the same distribution of the radargrams in $\mathcal{H}$, and ii) the discriminator D_H distinguishes between the translated samples $\mathcal{G}(\mathbf{R_L})$ and the real sample $\mathbf{R_H}$. Hence, the generator minimizes the loss function while the discriminator maximizes it. Similarly, for the Low2High GAN, we can define the adversarial loss function $L_{GAN}^{L}(\mathcal{F}, D_L, \mathcal{L}, \mathcal{H})$ to learn the generator mapping function $\mathcal{F}(\cdot)$ and the discriminator D_L. The cycle-consistency loss forces the CycleGAN to learn the features from the domains $\mathcal{L}$ and $\mathcal{H}$ and to perform domain translation [35]. Therefore, $L_{cyc}(G, F)$ forces the system to learn the two mapping functions $\mathcal{G} : \mathcal{L} \rightarrow \mathcal{H}$ and $\mathcal{F} : \mathcal{H} \rightarrow \mathcal{L}$ so that the translation is bidirectional. This is implemented by using the output of the Low2High GAN ($\widehat{\mathbf{R}}_H$) as input of the High2Low GAN and forcing the output of the High2Low GAN ($\widehat{\mathbf{R}}_L$) to be similar to original low-resolution radargram ($\widehat{\mathbf{R}}_L$). Therefore, the cycle consistency loss forces $\mathcal{G}(\cdot)$ to reverse $\mathcal{F}(\cdot)$, and therefore $\mathcal{F}(\mathcal{G}(\mathbf{R}_L)) \approx \mathbf{R}_L$ and $\mathcal{G}(\mathcal{F}(\mathbf{R}_H)) \approx \mathbf{R}_H$. The cycle consistency loss is defined as follows:

$$L_{cyc}(\mathcal{G}, \mathcal{F}) = E_{\mathbf{R}_L \sim pdata}(\mathbf{R}_L)[\|\mathcal{F}(\mathcal{G}(\mathbf{R}_L)) - \mathbf{R}_L\|] + E_{\mathbf{R}_H \sim pdata}(\mathbf{R}_H)[\|\mathcal{G}(\mathcal{F}(\mathbf{R}_H)) - \mathbf{R}_H\|] \quad (7)$$

The feature reconstruction loss L_{feat} is introduced to preserve the geometrical and radiometric details in the hidden-layer features and mitigate the low-pass effect introduced by the convolutions. The loss is inspired by [34] and, for each generator $i \in [L, H]$, compares the features extracted by the l-th encoder block ($H_c^{l,e}$) and the corresponding l-th decoder block ($H_c^{l,d}$) (see Fig. 8). Given that, $H_c^{l,e}$ contains more details than $H_c^{l,d}$, we train the model to reconstruct the details lost during the encoder compression and improve the geometrical details of the hidden-layer features. For each layer l, $H_c^{l,e}$ and $H_c^{l,d}$ have the same size. We compute a regression loss $L_l, l \in [1, N_L]$ that tends to be 0 when the features are similar, and thus the geometrical details are preserved. Here, we consider the mean square error (MSE) as it showed good performance with radar sounder data and for detail preservation [25], but any other regressive loss can be used:

$$L_l = \frac{1}{h_0 w_0} \sum_{r=1}^{h_0} \sum_{a=1}^{w_0} (H_c^{l,e} - H_c^{l,d})^2 \quad (8)$$

For each generator $i \in [L, H]$, we compute N_L feature reconstruction losses that are summed to obtain:

$$L_{feat}^{i} = \sum_{l=1}^{N_L} L_l^i \quad (9)$$

Hence, the total loss function L_{tot} is defined as a sum of all the losses:

$$L_{tot}(\mathcal{G}, \mathcal{F}, D_L, D_H) = \delta\, L_{cyc}(\mathcal{G}, \mathcal{F}) + \beta_H\, L_{GAN}^{H}(\mathcal{G}, D_H, \mathcal{L}, \mathcal{H}) + \beta_L\, L_{GAN}^{L}(\mathcal{F}, D_L, \mathcal{L}, \mathcal{H}) + \gamma_L\, L_{feat}^{L} + \gamma_H\, L_{feat}^{H} \quad (10)$$

where δ, β_H and β_L, and γ_L and γ_H control the relevance of the corresponding losses.

C. High-resolution radargram prediction

Once the CycleGAN is trained, the mapping function $\mathcal{G}$ learned by the Low2High generator can super-resolve previously unseen low-resolution radargrams. This is true if radargrams are acquired by the same pair of sensors used in the training and with a similar underlying geology. If needed, the mapping function $\mathcal{F}$ can reduce the resolution of unseen high-resolution radargrams.

V. Description of the Datasets

To assess the effectiveness of the proposed method, we consider a fully controlled experiment setup with the aim of super-resolving simulated radargrams. As high-resolution data, we use MCoRDS3 radargrams [3] from the Center for Remote Sensing of Ice Sheets (CReSIS) archives acquired over the polar areas of the Earth in Greenland and Antarctica. To conduct a controlled experiment, we generate low-resolution data using the analog-based simulator in [39] and the instrument parameters of the Earth-orbiting satellite (EORS) [38].

A. High-Resolution Radargram Pre-Processing.

For the MCoRDS3 Greenland dataset, the range and the azimuth resolutions are $\rho_r = 4.3$m and $\rho_a = 27.5$m, while for the Antarctica dataset, the range and the azimuth resolutions are $\rho_r = 2.6$m and $\rho_a = 27.5$m. The MCorDS3 data are pre-processed, including range compression, azimuth focusing with Synthetic Aperture Radar (SAR), and partial clutter compensation, resulting in resolutions outlined in Table I. The data are pre-processed by concatenating adjacent radargrams and removing the overlapping regions. We corrected the data for fluctuations resulting from undesired movements during data acquisition, such as pitch, roll, and yaw, which may have compromised the signal quality. As in [25], we normalize the power of the radargram based on the surface power for each column to reduce the fading of the power caused by the rolling movement of the aircraft on curves. We determined the maximum power for each azimuth column by utilizing the vector of the range coordinate of the surface $s(a)$. The power of the surface $P_s(a)$ for each azimuth trace was defined as $P_s(a) = \mathbf{R}(a, s(a))$. We masked the region above the surface to focus the network on studying the subsurface. The speckle noise affects the radar data, which can be approximated as multiplicative. To simplify the data handling and improve spatial information, we applied a log transformation to the power radargram to approximate the noise as additive.

B. Low-resolution Radargram Generation.

MCoRDS3 radargrams are processed with the simulator in [39, 6] and the parameters in [38] to generate the corresponding radargrams with a lower resolution. The simulator is based on reprocessing available radargrams (analog), considering differences between the analog and investigated scenario (e.g., instrument parameters and acquisition geometry) and their impact. The simulation approach i) extracts the geoelectrical model from the original radargram (i.e., high-resolution MCoRDS3 data) to estimate the target propagation loss to obtain the simulated radargram (i.e., low-resolution EORS data). The approach corrects ii) the signal amplitude by considering the RS orbital parameters, computing the orbital received power, and inputting the suitable parameter into the radar equation for a given SNR at the surface. iii) The approach also corrects for the noise as the two scenarios have different noise levels. The simulator stochastically adds the isotropic cosmic microwave background (CMB). The CMB noise power is modeled with the Rayleigh distribution [40]. Further, to reduce the range and azimuth resolutions to those expected by the orbiting acquisitions, the approach corrects iv) the bandwidth (narrower in the case of EORS) by applying a low-pass filter and v) the range and azimuth sampling according to the RS sampling frequency in range and the pulse repetition frequency (PRF) in azimuth by appropriately resampling the radargram. Further details can be found in [39, 6].

C. Dataset Setup

Each couple of high- and low-resolution radargrams is divided into patches, normalized by removing the mean and dividing by the standard deviation of the dataset. The couples are divided into three independent sets $\mathcal{J}^{tr}$, $\mathcal{J}^{val}$, and $\mathcal{J}^{ts}$ for the training, the validation, and the testing phases, respectively. We split the patches so that 80% of the couples are in $\mathcal{J}^{tr}$, 10% are in $\mathcal{J}^{val}$, and the remaining 10% in $\mathcal{J}^{ts}$. To assess the training robustness and avoid overfitting, we ensure that we do not include patches acquired in the same campaign in different sets ($\mathcal{J}^{tr}$, $\mathcal{J}^{val}$, and $\mathcal{J}^{ts}$).

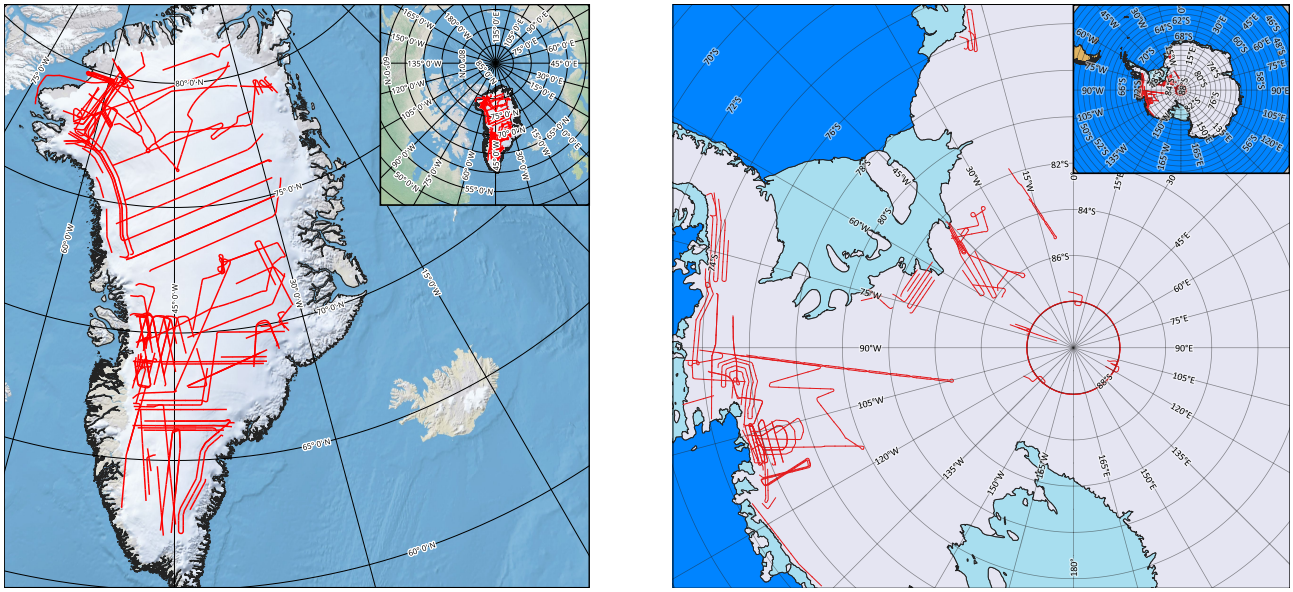
TABLE II: Dataset description.

Parameters	**Greenland dataset**	**Antarctica dataset**
Campaign number	38	38
Radargram number	908	626
Trace number	2724000	1878000
Total patch number	10640	7335
Training patch number ($\mathcal{J}_{tr}$)	8512	5868
Validation patch number ($\mathcal{J}_v$)	1064	733
Test patch number ($\mathcal{J}_{ts}$)	1064	734

Greenland Dataset. The high-resolution data for this dataset consists of 908 radargrams acquired over the inland of Greenland by MCoRDS3 [3] in 38 campaigns in 2013, 2017, and 2018. The details of the MCoRDS3 instrument and radargram properties are in Table I. The ground tracks of the campaigns are shown in Figure 9.a. The high- and low-resolution radargrams are divided into 10640 patches. The high-resolution patches have sizes $w = 256$ and $h = 1280$ in the azimuth and range directions, respectively. Low-resolution patches have sizes $w = 64$ and $h = 1920$. h is set so that the neural network has a complete view of the vertical distribution of the classes. The couples of corresponding high- and low-resolution patches are divided into the dataset $\mathcal{J}_{tr}$ of 8512 patches for the training phase, $\mathcal{J}_v$ of 1064 patches for the validation phase, and $\mathcal{J}_{ts}^1$ of 1064 patches for the testing phase (see Table II).

Antarctica Dataset. The high-resolution data for this dataset consists of 626 radargrams acquired over the inland of Antarctica by MCoRDS3 [3] in 38 campaigns in 2014, 2016, and 2018. Details of the MCoRDS3 instrument and the radargram are in Table I. The ground tracks of the campaigns are shown in Figure 9.b. High- and low-resolution radargrams are divided into 7335 patches; high-resolution patches have sizes $w = 256$ and $h = 1880$, and low-resolution patches have sizes $w = 64$ and $h = 1692$. The couples of corresponding high- and low-resolution patches are divided into $\mathcal{J}_{tr}$ of 5868 patches for the training phase, $\mathcal{J}_v$ of 733 patches for the validation phase, and $\mathcal{J}_{ts}^1$ of 734 patches for the test (see Table II).

Data Augmentation. Given the limited number of available training samples, data augmentation techniques, such as affine transformations and horizontal flips, are crucial to increase the dimension of the labeled dataset and improve the network invariance and robustness [41]. Similarly to [25], we use augmentation techniques that generate realistic augmented data that preserve the geometrical properties of geologic subsurface structures. More details can be found in the appendix A.



(a) (b)

Fig. 9: Ground tracks of the campaigns used for (a) Greenland and (b) Antarctica datasets.

TABLE III: Parameters of the proposed method and the neural network.

Parameter	Description	Greenland Dataset	Antarctica Dataset
(h_H, w_H)	High-resolution patch size in the range and the azimuth	(1280, 256)	(1880, 256)
(h_L, w_L)	High-resolution patch size in the range and the azimuth	(1920, 64)	(1692, 64)
M_0	Number of channels of the input data	1	
N_L	Number of downsampling and upsampling blocks	3	
$M_l, l = [1, \ldots, N_l]$	Features extracted by the l-th block of the encoder and the decoder	[64, 128, 256]	
$(k_a^l, k_r^l), l = [1, \ldots, N_l]$	Convolutional kernel size in the range and azimuth	[(5,4) (4,4) (4,4)]	[(3,1) (3,3) (3,3)]
$(k_x^l, k_y^l), l = [1, \ldots, N_l]$	Stride size in range and azimuth	[(1,2) (2,2) (2,2)]	[(1,2) (1,1) (1,1)]
ℓ_r	Learning rate	$5\,10^{-7}$	10^{-5}
$\mathcal{E}$	Maximum number of epochs for training phase	500	
N_B	Mini-batch size	32	

VI. Experimental Setup and Results

A. Experimental Setup

The parameters of the architecture are given in Table III. We used $N_L = 3$ downsampling blocks in the encoder and upsampling blocks in the decoder, and $N_R = 6$ residual blocks. The values of the kernels and strides are detailed in Table III. We train the model iteratively on batches of $N_B = 32$ patches to accelerate the network convergence by updating the parameters more frequently and reducing the variance of the updates. For the optimization, we used Adam optimization. The learning rate is set to $\ell_r = 5\,10^{-7}$ for the Greenland dataset, and $\ell_r = 10^{-5}$ for the Antarctica dataset after running several experiments varying the values in $[0.1, 10^{-9}]$. To enhance the generalization ability of the network and mitigate overfitting, the stopping criteria for the training phase rely on two factors– i) the maximum number of epochs, which is set to $\epsilon = 500$; and ii) the evaluation of the training and validation loss at each epoch.

To evaluate the efficacy of individual components of the proposed method, we perform an ablation study by comparing the outcomes of diverse experiments outlined in Table IV. The experimental setup gradually incorporates the essential elements of the proposed approach to determine their significance. Experiment 1 considers the basic network architecture without skip connections, attention gates, and ResNet modules trained with the feature reconstruction loss and without learning rate decay. Experiment 2 enriches the architecture in Exp 1 with the residual modules. Experiment 3 adds the skip connections and the learning rate decay after 250 epochs. Experiment 4 considers the network architecture without ResNet but with the attention gates and the learning rate decay after 75 epochs. Experiment 5 corresponds with the proposed method, with the attention gates and the ResNet modules, and the decay

TABLE IV: Description of the experimental setup.

Experiment	1	2	3	4	5
ResNet module	no	yes	yes	no	yes
Skip connections	no	no	yes	yes	yes
Attention gate	no	no	no	yes	yes
Feature reconstruction loss	yes	yes	yes	yes	yes
l_r decay (after xx epochs)	no	no	250	75	75

of the learning rate after 75 epochs. Finally, we compared the proposed method with the BWE in the literature [8].

B. Evaluation Criteria

For a quantitative evaluation of the proposed method effectiveness, we conducted a comparison between the original MCoRDS3 and the super-resolved patches in terms of signal-to-noise ratio (SNR), mean square error (MSE), peak signal-to-noise ratio (PSNR), signal-to-reconstruction error (SRE) in dB, structural similarity index (SSIM), and normalized cross-correlation coefficient (NCC). Reference values for SNR, PSNR, and SRE are those of the high-resolution data. For MSE, the reference value is 0, while for SSIM and NCC, it is 1. We refer to Appendix B for more details.

We validate the proposed method effectiveness by calculating the metrics over the whole patch (Case I), the upper part of the ice layers (green box in Fig. 1, Case II) and on the basal region (yellow box in Fig. 1, Case III). The choice of these regions is driven by the analysis to understand the impact of the artifacts in the super-resolved patches. The results in [10] and the log-ratio images between the original and the super-resolved patches (see Fig. 10.i and 14.i) that show that the main artifacts are in the ice stratigraphy and the basal interface.

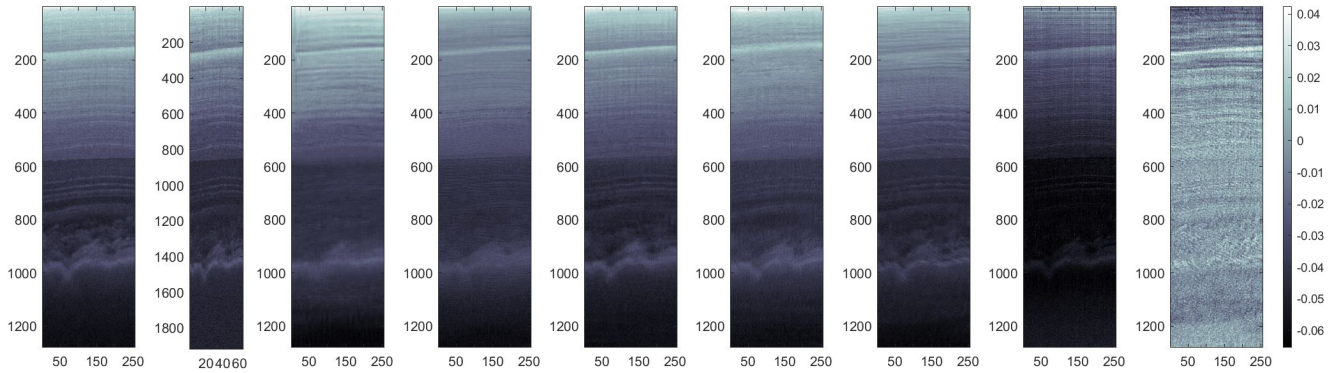
C. Experimental Results on the Greenland Dataset

Fig. 10, 11, and 12 show the qualitative results by varying the experimental setup as defined in Table IV. Radargrams are selected to represent the complexity of the datasets. We focus on radargrams showing reflections with complex shapes, such as diffuse and irregular reflections. The qualitative results show that the proposed method (Exp 5) generates visually sharper super-resolved patches than the others (Exp 1-4) and the literature BWE method [8], confirming the importance of each element of the architecture and the loss design. The proposed method super-resolves the complex ice stratigraphy in the upper part of the ice sheet with good accuracy (see Fig. 11), dramatically improving the smoother results of [10]. Moreover, the proposed method (column (g) in Fig. 10 and 11), having a more complex architecture than that in Experiments 1-4, generates an ice layering that is visually and radiometrically more realistic (columns (c)-(f)). The upper layering in Exp 1 and 4 shows unrealistic reflections, which improve in Experiments 2, 3, and 5 using the ResNet module at the bottleneck. Moreover, skip connections (Exp 1-2 vs. Exp 3-5) generate sharper and more detailed patches. The proposed method also shows sharper reflections than the BWE method [8]– column (h) in Fig. 10 and 11. This is also true for the bottom part of the ice sheet and the basal interface (see Fig. 12). This area shows irregular reflections that follow the bedrock topography. Note the improvement in sharpness in the super-resolution caused by the ResNet module (Exp 1,4 vs. Exp 2,3,5) and skip connections (Exp 1,2 vs. Exp 3-5). The proposed method correctly super-resolves the topography of the bedrock, generating fewer artifacts than in Experiments 1-4 and BWE. Moreover, the basal ice at the ice core bottom, appearing as a diffuse reflection, is super-resolved with higher visual precision by the proposed method than in Experiments 1-4. Further, proper scheduling for the learning rate helps train the network better and achieves super-resolved patches with higher details (Exp 1-3 vs. Exp 4-5). The artifacts generated by the proposed method are visible in ice layering in Fig. 10.h, which shows the stretched log-ratio of the high-resolution and Exp 5 super-resolved patches, thus highlighting where the super-resolved patch does not adhere to the original high-resolution one. The artifacts are mostly related to the sharpness of the reflections and not to the presence of unrealistic reflections. This is not true for the BWE method results [8], as it generates unrealistic reflections in the ice layering (see column (h) in Fig. 11). The goodness of the proposed method is also visible in Fig. 13 that plots the average rangeline for the high-resolution patch in blue and super-resolved patches with the proposed method (Fig. 10) in red. The average rangeline of Exp 5 shows to follow the trend and the peaks of the original high-resolution ones. Whereas the average rangeline of the super-resolved patch with the BWE method (in yellow) [8] shows large fluctuations that do not follow the high-resolution one.

Table V reports the quantitative results of Experiments 1-5 and the BWE method [8] calculated over the whole patch and thus depth (Case I), and in the regions with reflections complex to super-resolve: the ice layers (Case II) and the basal region (Case III). The proposed method outperforms Experiments 1-4, and the BWE method [8], highlighting the importance of each element of the proposed architecture. For Case I (metrics calculated over all the patch depth), the SNR and PSNR values are higher or closer to the proposed method's reference values, equal to 5.36 dB and 39.69 dB, respectively. The SNR and PSNR values for the other Experiments are lower, up to about 2 dB and 4 dB (Experiment 1), respectively. The proposed method achieves a more accurate SRE than Experiments 1-4. Exp 5 has an SRE value equal to 8.89 dB, while for the other experiments, it is lower and in the range between 6.49 dB and 7.32 dB. The SSIM and NCC metrics indicate a good structural correlation between the high-resolution MCoRDS data and the super-resolved patches. In particular, for the proposed method (Exp 5), the SSIM index equals 0.89, and the NCC equals 0.9878. The values of the metrics calculated for the complex regions in Case II (upper part of the ice layers) and III (basal region) confirm the quality of the proposed method (see Table V). The average values of the MSE, SSMI, and the NCC, being $1.8e^{-2}$, 0.9131 and 0.9851, respectively, indicate a high correlation between the upper part of the layer and the basal areas of the high-resolution patch and the super-resolved one.

D. Experimental Results on the Antarctica Dataset

Fig. 14 shows the qualitative results for the Antarctica dataset. The results show that the proposed method accurately super-resolves the radargrams and confirm the significance of all the elements of the network architecture and the training function. Fig. 14 shows that enriching the network with the ResNet, attention, and skip connections improves the quality of the horizontal ice stratigraphy in the upper part of the patches and the basal interface area, that have reflections with a complex shape. Moreover, we (Exp 5) significantly reduced



(a) H-Res (b) L-Res (c) Exp 1 (d) Exp 2 (e) Exp 3 (f) Exp 4 (g) Exp 5 (h) BWE (i) LR

Fig. 10: Results for the Greenland dataset– (a) MCoRDS3 high-resolution patch, (b) low-resolution simulated patch [6], (c)-(g) super-resolved patches for Exp 1-5, (h) BWE super-resolved patch [8], and (i) log-ratio (LR) between (a) and (g).

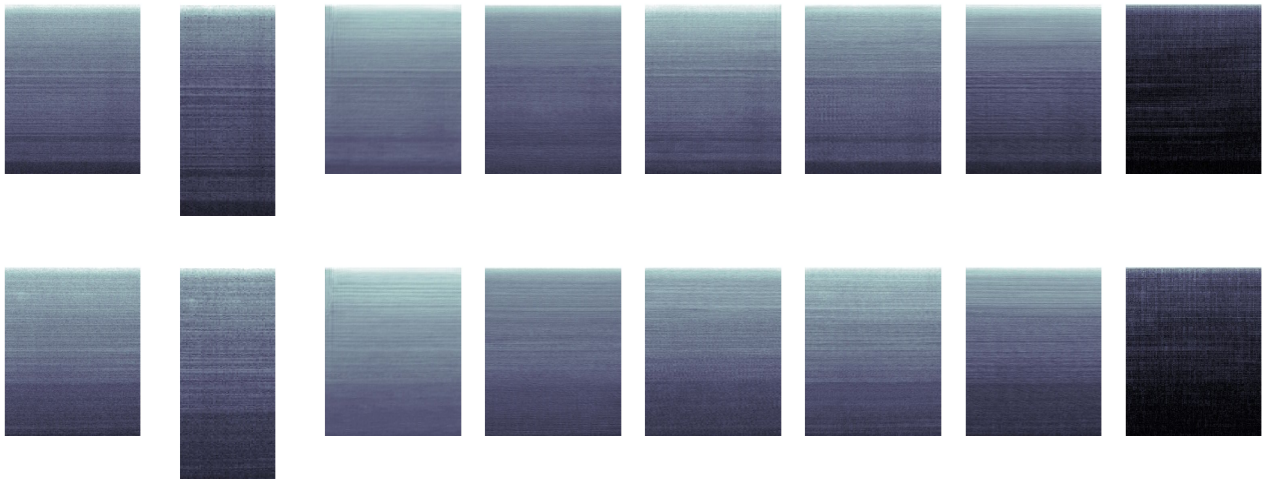
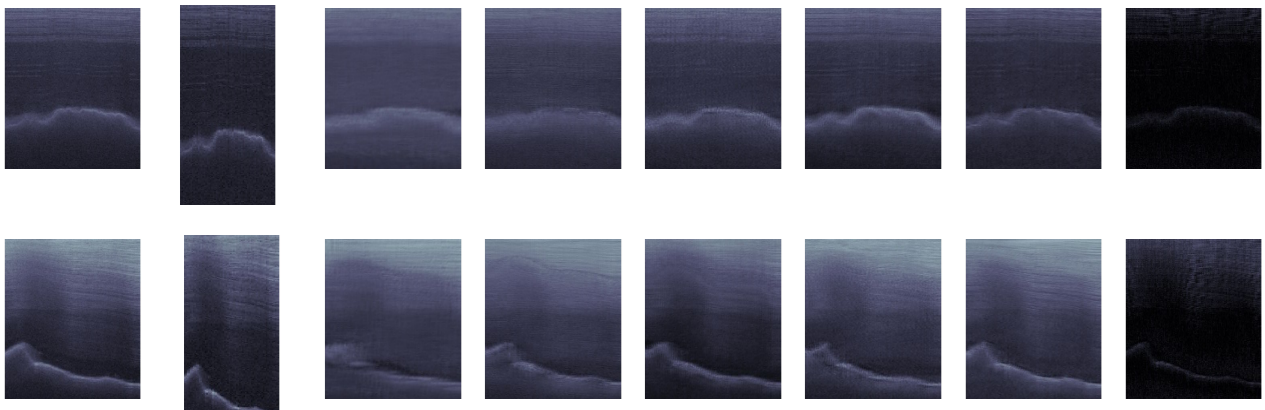


Fig. 11: Details of patches highlighting Case II region for the Greenland dataset– (a) MCoRDS3 high-resolution patch, (b) low-resolution simulated patch [6], (c)-(g) super-resolved patches for Experiments 1-5, and (h) super-resolved patches with BWE [8].



(a) High Res (b) Low Res (c) Exp 1 (d) Exp 2 (e) Exp 3 (f) Exp 4 (g) Exp 5 (h) BWE

Fig. 12: Details of patches highlighting Case III region for the Greenland dataset– (a) MCoRDS3 high-resolution patch, (b) low-resolution patch simulated with [6], (c)-(g) super-resolved patches for Experiments 1-5, and (h) super-resolved patches with BWE [8].

TABLE V: Quantitative performance of the proposed super-resolving method on the Greenland dataset.

Patch	Metrics		Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	BWE [8]	Ideal Value
Case I	**SNR (dB)**	mean	3.8095	4.3621	4.4023	4.9417	5.3621	1.2674	5.3164
		std	0.6668	0.7046	0.6402	0.5566	0.4087	–	0
	MSE	mean	$1.86e^{-2}$	$1.70e^{-2}$	$1.64e^{-2}$	$1.54e^{-2}$	$1.49e^{-2}$	0.141	0
		std	0.1232 e^{-2}	0.1161 e^{-2}	0.1419 e^{-2}	0.2158 e^{-2}	0.1047 e^{-2}	–	0
	PSNR (dB)	mean	35.5168	36.2513	37.8062	38.4969	39.6965	18.4972	-
		std	1.9907	1.6439	1.7327	1.8167	1.9309	–	0
	SRE (dB)	mean	6.4911	6.7782	6.8155	7.3193	8.9848	2.5393	-
		std	0.2861	0.2769	0.4814	0.3359	0.2677	–	0
	SSMI	mean	0.8267	0.8312	0.8650	0.8897	0.8909	0.1629	1
		std	0.0528	0.0561	0.0533	0.0528	0.0513	–	0
	NCC	mean	0.9329	0.9648	0.9672	0.9762	0.9878	0.2126	1
		std	0.0526	0.0537	0.0457	0.0404	0.0463	–	0
Case II - ice layers	**SNR (dB)**	mean	4.9333	5.1260	5.4471	5.8954	6.0962	2.6008	6.0452
		std	0.8687	0.2880	0.2338	0.1369	0.1647	–	0
	MSE	mean	$2.40e^{-2}$	$2.23e^{-2}$	$2.36e^{-2}$	$1.85e^{-2}$	$1.54e^{-2}$	0.1620	0
		std	0.1588 e^{-2}	0.4361 e^{-2}	0.2388 e^{-2}	0.1923 e^{-2}	0.4331 e^{-2}	–	0
	PSNR (dB)	mean	36.7571	36.7814	36.9062	37.8062	38.1312	20.4732	-
		std	2.3167	2.0861	1.9415	1.9916	1.6875	–	0
	SRE (dB)	mean	9.4467	9.6286	9.7731	9.5521	10.3732	–	-
		std	0.4214	0.4372	0.5708	0.5812	0.3329	2.7825	0
	SSMI	mean	0.8506	0.8522	0.8752	0.8742	0.8884	0.4195	1
		std	0.0555	0.0728	0.0684	0.0572	0.0564	–	0
	NCC	mean	0.9451	0.9564	0.9531	00.9735	0.9872	0.5966	1
		std	0.0738	0.0761	0.0643	0.0649	0.0716	–	0
Case III - basal region	**SNR (dB)**	mean	3.9333	4.3610	4.6787	4.9460	5.1915	2.3897	5.272
		std	0.9687	0.4396	0.5834	0.3543	0.3999	–	0
	MSE	mean	$2.63e^{-2}$	$2.44e^{-2}$	$2.34e^{-2}$	$2.26e^{-2}$	$1.85e^{-2}$	0.1659	0
		std	0.1588 e^{-2}	0.1649 e^{-2}	0.1417 e^{-2}	0.1361 e^{-2}	0.1621 e^{-2}	–	0
	PSNR (dB)	mean	36.3758	36.7846	36.7405	37.0307	38.1312	17.8343	-
		std	2.0829	2.1231	2.0918	1.9167	1.8810	–	0
	SRE (dB)	mean	9.7865	9.9406	10.2864	10.0640	10.3732	2.1953	-
		std	0.5812	0.4395	0.6168	0.6889	0.3783	–	0
	SSMI	mean	0.8884	0.8839	0.8993	0.8996	0.9131	0.3734	1
		std	0.0555	0.0424	0.0436	0.0320	0.0343	–	0
	NCC	mean	0.9151	0.95436	0.9560	0.9762	0.9851	0.3755	1
		std	0.0738	0.0762	0.0651	0.0712	0.0657	–	0

the artifacts and improved the details compared to Exp 1-4 and the BWE method. Finally, it is worth noticing that most of the artifacts introduced by the proposed method are related to the sharpness of the reflections (see 14. i) and not to the presence of unrealistic reflections.

Similar to the Greenland dataset, the quantitative results confirm the proposed method effectiveness in super-resolving radargrams with complex reflections (see Table VI). For Case I, the SNR, PSNR, and SRE values of Exp 5 are the closest to the ideal value and increase by about 1, 9, and 5 dB, respectively, compared with Exp 1. The MSE value decreases from 0.039 of Exp1 to 0.011 Exp 5, while the SSMI and NCC values (reference value 1) increase from 0.75 and 0.68 of Exp 1 to 0.86 and 0.98 of Exp 5, respectively. This trend is also visible in Cases II and III– the SNR, SRE, and PSNR values increase by at least 0.7 dB in Exp 5 compared with Exp 1. Also, the SSMI and NCC values increase, highlighting the method effectiveness in super-resolving complex reflections, such as those of the ice stratigraphy and the basal interface.

VII. Discussion and Conclusions

This study fills a significant gap in the existing literature by presenting the first method for super-resolving radargrams in the range and azimuth directions simultaneously. The method exploits the capability of generative models to learn the relation between the radar signals and the geology of a given area at different scales. The method, in terms of problem formulation, architecture, and training design, is tailored to the properties of radargrams with different resolutions. The ablation analysis confirms the significance of each system element (ResNet module, skip connection, attention mechanism, and scheduled learning rate) as Exp 5 performs better than Exp

TABLE VI: Quantitative performance of the proposed super-resolving method on the Antarctica dataset.

Patch	Metrics		Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	BWE [8]	Ideal Value
Case I	**SNR (dB)**	mean	4.5227	4.6469	4.9095	4.9112	5.0799	2.2674	5.1466
		std	0.4212	0.7883	0.9400	0.8351	0.4980	–	0
	MSE	mean	0.0393	0.0257	0.0252	0.0154	0.0108	0.148	0
		std	0.0038	0.0028	0.0029	0.0085	0.0011	–	0
	PSNR (dB)	mean	19.5280	20.8770	26.4867	27.9605	28.0194	9.1893	-
		std	14.6399	1.8013	1.8753	1.2673	1.8166	–	0
	SRE (dB)	mean	1.7730	2.5294	5.6006	6.6572	6.6681	1.2862	-
		std	0.2697	0.2325	0.2375	0.17898	0.2094	–	0
	SSMI	mean	0.7490	0.8053	0.8151	0.8381	0.8601	0.1948	1
		std	0.0670	0.0549	0.0643	0.0883	0.1022	–	0
	NCC	mean	0.6800	0.7782	0.9470	0.9714	0.9853	0.3265	1
		std	0.1246	0.1744	0.1476	0.0540	0.0553	–	0
Case II - ice layers	**SNR (dB)**	mean	4.7602	4.6970	4.9514	5.0827	5.2921	2.0249	5.1577
		std	0.4017	0.3354	0.2992	0.3104	0.2520	–	0
	MSE	mean	0.0968	0.0745	0.0732	0.0331	0.0523	0.1623	0
		std	0.0011	0.0011	0.0011	0.0032	0.0042	–	0
	PSNR (dB)	mean	14.9200	16.8978	24.3845	25.5104	25.7761	11.0527	-
		std	1.5456	1.8430	1.9050	1.6318	1.0178	–	0
	SRE (dB)	mean	1.3376	2.3694	5.9117	7.3139	7.4315	0.9529	-
		std	0.3630	0.3427	0.3325	0.2408	0.2626	–	0
	SSMI	mean	0.4823	0.5822	0.6909	0.7053	0.7546	0.5620	1
		std	0.1007	0.1140	0.1206	0.1311	0.1175	–	0
	NCC	mean	0.5748	0.7244	0.9050	0.9595	0.9810	0.4966	1
		std	0.1554	0.0818	0.0991	0.1452	0.0584	–	0
Case III - basal region	**SNR (dB)**	mean	3.1693	3.1983	3.2872	3.2556	4.6274	2.3897	4.8170
		std	0.6612	0.5777	0.8659	0.7238	0.2921	–	0
	MSE	mean	0.0050	0.0082	0.0060	0.0037	0.0036	0.1111	0
		std	0.0008	0.0007	0.0008	0.0002	0.0002	–	0
	PSNR (dB)	mean	25.0596	25.0165	30.6781	31.0698	31.7145	17.2395	-
		std	2.7965	2.5262	2.3314	2.5186	2.5975	–	0
	SRE (dB)	mean	3.2071	3.1376	6.4768	6.6639	7.5281	2.5945	-
		std	1.0054	0.9649	0.8654	0.8931	0.5050	–	0
	SSMI	mean	0.7573	0.7597	0.8303	0.8600	0.9109	0.2906	1
		std	0.0974	0.0760	0.0898	0.0126	0.0120	–	0
	NCC	mean	0.8546	0.8723	0.9361	0.9866	0.9644	0.2385	1
		std	0.0167	0.0126	0.0204	0.0232	0.0248	–	0

1-4 and the BWE literature method [8] in terms of average SRE, MSE, NCC, and SSMI. Further, the qualitative analysis shows that the proposed method accurately generates super-resolved radargrams (Case I) and reproduces the complex reflections (Case II- ice stratigraphy, and Case III- basal area and interface) by sharpening the low-resolution details without introducing significant distortion and artifacts. Finally, the proposed method accurately super-resolves low-resolution radargrams with the resolution in [6] to the different range resolutions in the ice of Greenland and the Antarctica datasets (4.3 m and 2.6 m, respectively).

Given that the network learns to interpret the low-resolution signal based on the geological and statistical properties of the training data, the network super-resolves in a reliable way only radargrams acquired by the same sensors and with similar geological conditions as the training data. For data acquired by instruments with different parameters, as future Earth-orbiting sounders, like STRATUS [38] and DEBRIS [42], the deep model is expected to perform well after fine-tuning or training from scratch. Fine-tuning is feasible with a limited number of epochs and samples, as demonstrated in recent works for RS data segmentation and denoising [25, 43, 44]. Data acquired by RS with different parameters or in areas with different geologic conditions (e.g., desert or arid areas), training from scratch is suggested with a comprehensive dataset (data acquired with the low- and high-resolution instruments over the same areas with the same geologic conditions), even if it requires more computational time. As a reference, in our experiments on an NVIDIA GeForce RTX 3090 GPU, training took approximately 12 hours (500 epochs). However, once trained, the model performs rapid super-resolution (less than 2 seconds per mini-batch).

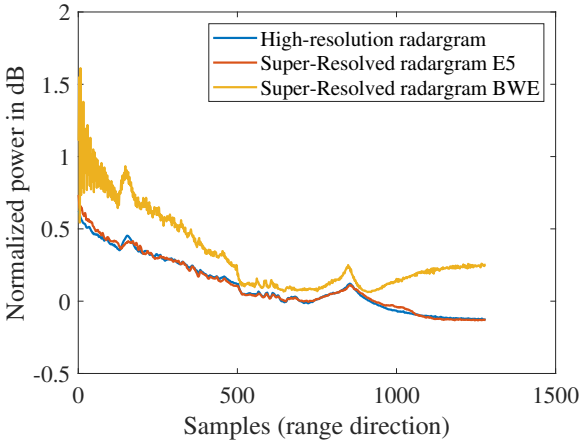


Fig. 13: Average rangelines for the patch in Fig. 10 super-resolved with the proposed method (Exp 5) and BWE [8].

To reduce the computational time and the size of the training dataset, as done in the RS literature [43, 25, 44], domain adoption and transfer learning techniques can be used to pre-train the model on a source task that can be different from the destination task (e.g., classification as source task and segmentation as destination task) and a source dataset different from RS, such as computer vision datasets. Given catastrophic forgetting, pre-training on RS data with a source task unrelated to the destination task (such as radargram reconstruction) facilitates the subsequent fine-tuning with fewer epochs and samples [25, 44]. If the fine-tuning is robust, the pre-training can be done with a task different from super-resolution and not-RS datasets, as well as with the super-resolution task and data acquired by different sensors imaging a different geology.

A limitation of the proposed method is the potential presence of artifacts in the super-resolved radargrams (albeit significantly less than observed in our prior work [10] and the BWE method [8]). Some artifacts should be expected as the super-resolved radargrams are estimated with a mapping function learned by the network as a solution to an ill-posed problem. Hence, artifacts originate from the network challenge in generalizing the properties of low-resolution data and accurately translating them to higher resolution. In our earlier work [10] (Experiment 2 in this manuscript), we highlighted key challenges in super-resolving with a deep model, such as the lack of sharpness (e.g., a low-pass effect in ice stratigraphy and diffuse reflections at the basal interface) and the presence of non-realistic artifacts in reflections. Despite these challenges, our architecture and training loss actively mitigate artifacts and enhance the overall quality of the super-resolved data. Uncommon buried structures, changes in ice dynamics not adequately captured, or infrequent occurrences may result in less accurate super-resolution. This is especially true for significant climate-induced alterations in freeze-thaw cycles impacting layering structures. These patterns should be accurately modeled in the training dataset to mitigate the presence of artifacts. Under this assumption, our proposed method promises to super-resolve them correctly.

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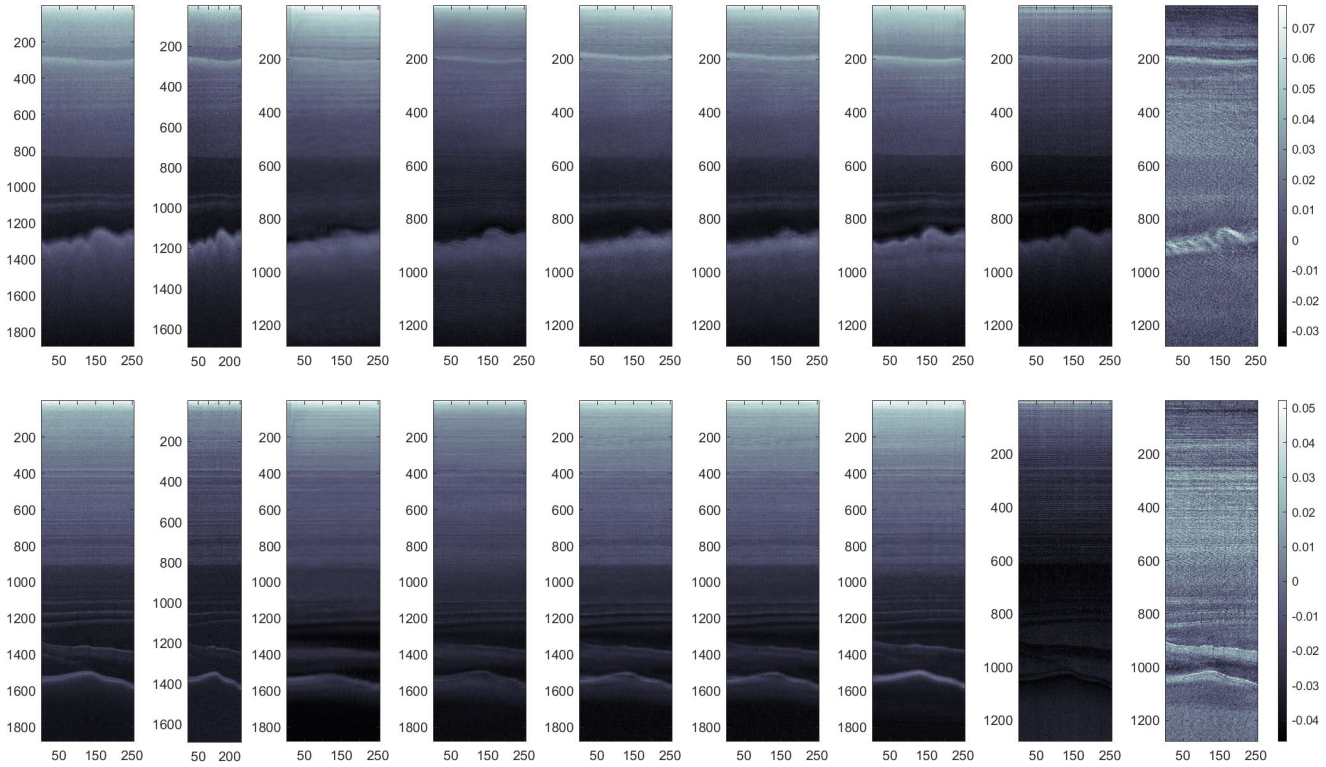
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(a) HRes (b) LRes (c) Exp 1 (d) Exp 2 (e) Exp 3 (f) Exp 4 (g) Exp 5 (h) BWE (i) LR

Fig. 14: Results for the Antarctica dataset– (a) MCoRDS3 high-resolution patch, (b) low-resolution simulated patch [6], (c)-(g) super-resolved patches for Exp 1-5, (h) BWE super-resolved patch [8], and (i) log-ratio (LR) between (a) and (g).

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APPENDIX A
DATA AUGMENTATION

As data augmentation, we apply random horizontal flips, rotation, and elastic deformations to the high- and low-resolution

patches to generate new pairs. The probability of applying a random horizontal flip is set to $p_f = 0.5$. The probability of applying a random rotation is set to $p_r = 0.8$, with rotation degrees in the range of $[-\theta_{max}, +\theta_{max}]$, where θ_{max} is the maximum steepness angle of the surface. Furthermore, a random elastic deformation transformation is applied to the vertical and horizontal directions with a $p_e = 0.9$ probability (Δ_r and Δ_a, respectively). The elastic deformation transformation uses a random displacement field based on coarse grids of dimensions 4×4, 8×8, and 16×16 cells. For each cell and direction, the displacement vectors are sampled from a Gaussian distribution with standard deviation σ_e, where σ_e is sampled from another Gaussian distribution with mean $\mu'_e = 8$ and standard deviation $\sigma'_e = 0.6$. Using the stress field on the patches involves moving each pixel to a new position and using spline interpolation of order one, as explained in [46]. A mirroring resampling filter mitigates the low-pass effect at the border pixels. The parameters of the Gaussian distribution are chosen experimentally to ensure that the deformations in the augmented patches appear realistic.

APPENDIX B
EVALUATION METRICS

The SNR in dB is separately calculated for the original and super-resolved patches and is defined for a given patch $P(r, a)$ as:

$$\text{SNR}_{\text{dB}}(P) = P_{\text{signal,dB}} - P_{\text{noise,dB}} \tag{11}$$

where $P_{\text{signal,dB}}$ and $P_{\text{noise,dB}}$ are the signal and noise power levels. They are defined as the root-sum-of-squares levels of the signal and the noise, respectively, in the radargram– $P_{\text{signal,dB}} = \sqrt{\sum_{r,a} |R_{\text{signal}}(r, a)|^2}$, where R_{signal} indicates the signal areas of the patch. Similarly, $P_{\text{noise,dB}} = \sqrt{\sum_{r,a} |R_{\text{noise}}(r, a)|^2}$ is the noise level, where R_{noise} are the noisy areas of the patch (e.g., above the surface or below the last return).

The MSE is defined as the mean of the squares of the errors between the original and super-resolved patches:

$$\text{MSE}(P_t, P_{\text{SR}}) = \frac{1}{wh} \sum_{r=1}^{h} \sum_{a=1}^{w} (P_t(r, a) - P_{\text{SR}}(r, a))^2. \tag{12}$$

MSE tends to 0 when $P_t(r, a) - P_{\text{SR}}(r, a)$ are similar.

The PSNR in dB indicates the grade of reconstruction between the original and the super-resolved patches and is defined as:

$$\begin{aligned} \text{PSNR}_{\text{dB}}(P_t, P_{\text{SR}}) = 20 \log(\text{MAX}(P_t)) \\ -10 \log(\text{MSE}(P_t, P_{\text{SR}})) \end{aligned} \tag{13}$$

where $\text{MAX}(\cdot)$ indicates the maximum value of the patch P_t. The SRE in dB is the ratio of the power of the original image to the power of the error introduced by the super-resolution process and is defined as:

$$\text{SRE}(P_t, P_{\text{SR}}) = 10 \log \frac{||P_t||_2^2}{||P_t - P_{\text{SR}}||_2^2}. \tag{14}$$

A higher SRE in dB indicates a better super-resolution quality, as the difference between P_t and P_{SR} is smaller relative to the power of the original image.

SSIM index evaluates the similarity between two patches by measuring the perceptual difference between the radargrams based on their intensity, contrast, and structural information. The similarity score is between 0 and 1, with 1 indicating perfect similarity. The SSIM is defined as:

$$\begin{aligned} \text{SSIM}(P_t, P_{\text{SR}}) = [L(P_t, P_{\text{SR}})]^{\xi} + [C(P_t, P_{\text{SR}})]^{\zeta} + \\ [S(P_t, P_{\text{SR}})]^{\iota} \end{aligned} \tag{15}$$

where $L(P_t, P_{\text{SR}})$ is the intensity similarity between P_t, P_{SR}, $C(P_t, P_{\text{SR}})$ is the contrast similarity between P_t and P_{SR}, and $S(P_t, P_{\text{SR}})$ is the structure similarity between P_t and P_{SR}. Finally, ξ, ζ, and ι are weighting parameters that control the relative importance of the luminance, contrast, and structure similarities, respectively.

The NCC coefficient measures the similarity between P_t and P_{SR} by calculating the correlation between the patches and is defined as:

$$\text{NCC}(P_t, P_{\text{SR}}) = \frac{\sum_{r,a} P_t(r, a) P_{\text{SR}}(r, a)}{\sqrt{\sum_{r,a} (P_t(r, a) - \mu_t)^2} \sqrt{\sum_{r,a} (P_{\text{SR}}(r, a) - \mu_{\text{SR}})^2}} \tag{16}$$

The NCC value ranges between -1 and 1, with 1 indicating perfect similarity between the two images, 0 indicating no correlation, and -1 indicating perfect anti-correlation or dissimilarity.